AI.210

Deep Learning

COURSE OVERVIEW

Deep neural networks: CNNs, RNNs, transformers, and modern architectures for perception and sequence tasks.



ILLUSTRATIVE HERO IMAGERY

DEEP LEARNING

1

SESSION

60 MINUTES

ORIENTATION: THE DEEP REVOLUTION

KEY TOPICS

- From shallow to deep
- Why depth matters
- Data and compute needs
- Deep learning in AI and robotics

TEACHING FOCUS

Enter the era of deep representations.

▲ **KEY TAKEAWAY** Depth learns hierarchy.

DEEP LEARNING

2

SESSION

60 MINUTES

CONVOLUTIONAL NETWORKS

KEY TOPICS

- Convolution operation
- CNN architectures
- Pooling and normalization
- Transfer learning

TEACHING FOCUS

Process images and spatial data.

KEY TAKEAWAY CNNs see patterns across space.

3

SESSION

60 MINUTES

RECURRENT NETWORKS

KEY TOPICS

- RNNs and LSTMs
- Sequence modeling
- GRUs
- Time-series forecasting

TEACHING FOCUS

Handle data that unfolds over time.

KEY TAKEAWAY Sequences carry memory.

4

SESSION

60 MINUTES

ATTENTION & TRANSFORMERS

KEY TOPICS

- Attention mechanism
- Transformer architecture
- Self-attention
- Language and vision transformers

TEACHING FOCUS

Focus on what matters.

KEY TAKEAWAY Attention weights relevance dynamically.

5

SESSION

60 MINUTES

TRAINING DEEP NETWORKS

KEY TOPICS

- Backprop through depth
- Optimizers: Adam, RMSprop
- Batch normalization
- Regularization

TEACHING FOCUS

Train very deep models reliably.

KEY TAKEAWAY Depth brings power and training challenges.

6

SESSION

60 MINUTES

DEEP LEARNING FOR ROBOTS & REVIEW

KEY TOPICS

- Deep perception and control
- Deployment on edge
- Course recap and assessment
- Toward pattern recognition (AI.211)

TEACHING FOCUS

Deploy deep models on embodied agents.

KEY TAKEAWAY Deep learning gives robots rich perception.